

Answer: B

10 Loss in gravitational PE =  $mgh = 4.2 \times 9.81 \times 2.9 = 12 \text{ J}$ .

Gain in gravitational PE =  $mgh = -12 \text{ J}$ .

Gain elastic PE =  $\frac{1}{2} mgh = +6 \text{ J}$

Answer: B

11  $\omega = \frac{2\pi}{T} = \frac{2\pi}{10^{-7}} = 1.99 \times 10^7 \text{ rad s}^{-1}$ .